

Application of the Population Health Institute Model of Health for Identifying Cancer Catchment Area Priorities

Amy Trentham-Dietz, PhD^{1,2} ; Thomas P. Lawler, PhD¹ ; Ronald E. Gangnon, PhD^{1,2,3} ; Allison R. Dahlke, MPH¹ ; Noelle K. LoConte, MD, FASCO^{1,4} ; Earlise C. Ward, PhD^{1,5} ; Christine P. Muganda, PhD⁶ ; Shaneda Warren Andersen, PhD^{1,2} ; and Marjory L. Givens, PhD^{2,6}

DOI <https://doi.org/10.1200/CCI-25-00126>

ABSTRACT

PURPOSE The University of Wisconsin Population Health Institute (PHI) Model of Health, grounded in models developed over a decade ago, provides a framework for prioritizing health-related investments including setting agendas, implementing policies, and sharing resources for improving community health and health equity. The model includes multiple determinants of health and two broad health outcomes (length and quality of life). We adapted the PHI Model of Health to cancer outcomes.

METHODS Using county-level publicly available data, health factor summary measures were derived in three areas: health infrastructure including health promotion and clinical care, physical environment, and social and economic factors. A composite health factor z-score was calculated as the weighted (40%, 15%, and 45%, respectively) average of the summary measures for each county, and k-means clustering was used to create unequally sized county groups with lower (healthier) to higher (less healthy) z-scores. We fit age-adjusted negative binomial regression models to estimate rate ratios and 95% CI for cancer mortality in relation to county health factor cluster.

RESULTS Age-adjusted cancer mortality rates increased across the 10 county health factor clusters for all-cancers as well as for lung, colorectal, breast, and prostate cancers. Rate ratios generally increased across the 10 health factor clusters for all cancers combined and for specific cancer types. Compared with counties with the most favorable health factor conditions, the counties with the least favorable conditions had an all-cancer mortality rate ratio of 1.49 (95% CI, 1.39 to 1.60).

CONCLUSION The PHI model of health adapted to cancer outcomes provides an approach for linking community-specific conditions to the interventions that hold promise to directly address drivers of the cancer burden.

ACCOMPANYING CONTENT

 Appendix

Accepted December 22, 2025

Published February 5, 2026

JCO Clin Cancer Inform

10:e2500126

© 2026 by American Society of
Clinical Oncology

INTRODUCTION

National Cancer Institute (NCI)–designated cancer centers must identify a geographic area, referred to as a catchment area, that each center serves in its research, clinical care, education, and community outreach and engagement.¹ This requirement is intended to focus the center's mission toward reducing the cancer burden for the catchment area's population.

Centers have used different approaches to identify priorities. Often, centers use a combination of sources of information in these decisions, including cancer incidence and mortality rates, prevalence of risk factors, and cancer screening rates within the catchment area; knowledge about the areas of

scientific and clinical expertise and strengths at their centers; and input from community members. Opportunities remain to specify a framework for decision making that takes a systems science approach including consideration of multiple possible interacting, heterogeneous, and dynamic streams of data for different populations and geographic regions within catchment areas.² Effective cancer prevention and control efforts need to leverage data for intervening at multiple levels—patient, provider, health system, and societal—with multiple complementary approaches.

The University of Wisconsin Population Health Institute (PHI) Model of Health³ (an evolution formerly known as the County Health Rankings Model of Health⁴) offers promise as a rigorous approach for identifying catchment area priorities

CONTEXT

Key Objective

Can the Population Health Institute Model of Health provide a data framework for informing decision making to address health-related disparities within cancer center catchment areas through systems, policy, and public health interventions?

Knowledge Generated

Using the model framework to cluster counties with similar community conditions describing intervenable health factors, most health factors sorted in a dose-response manner across the clusters, for example, access to screening tests, exercise opportunities, broadband Internet, insurance, and education. The county cluster with the least favorable community conditions had a 49% greater all-cancer mortality rate ratio than the county cluster with the most favorable conditions.

Relevance (*M. Behera*)

This model provides a structured framework for linking community-level determinants, such as socioeconomic factors, access to care, and environmental conditions, to actionable interventions that reduce cancer burden. This model enables health systems and policymakers to prioritize investments that improve both length and quality of life for populations disproportionately affected by cancer.*

*Relevance section written by JCO CCI Associate Editor Madhusmita Behera, PhD.

for all NCI-designated cancer centers. Supported by the Robert Wood Johnson Foundation, the PHI Model of Health has been used for over two decades to aid health professionals and decision makers in understanding community conditions favorable to living long and well and in identifying evidence-informed approaches for improving community health and health equity. The model draws on a theory of change and recognition of the cross-sector participation needed to catalyze community health improvement. The model draws attention to factors beyond clinical care and individual behavior change that shape community health.³ The PHI Model of Health has had national impact, yet has generally been defined and implemented to address broad population health outcomes of quality and length of life.

In this report, we adapt the PHI Model of Health to cancer for counties across the United States and demonstrate the strengths of this approach for identifying priorities in an NCI-designated cancer center catchment area. Specifically, the model is used to identify counties with risk factors that may be driving higher cancer death rates, which may then be prioritized as targets for intervention.

METHODS

To adapt the PHI Model of Health to cancer, we modified the health outcomes (population health and well-being) and health factors (community conditions) while retaining the main model structure including county as the geographic unit. We then compared cancer outcomes with a composite index of the health factors, divided into clusters, to demonstrate variation across >3,000 counties in the United States.

Health Outcomes: County-Level Cancer Mortality Rates

The PHI Model of Health includes length and quality of life as health outcomes. For this analysis, we instead used county-level cancer mortality rates from the Centers for Disease Control and Prevention (CDC) State Cancer Profiles website.⁵ Cancers of interest included all cancers combined and lung, colorectal, female breast, and prostate cancers separately. Data for counties with an average of three or fewer cases per year for a given outcome were suppressed, and consequently, these counties were excluded from the analysis. Data from 3,079 of 3,143 (98.0%) US counties were available for all-cancer mortality.

Health Factors: County-Level Community Conditions

Health factors in the PHI Model of Health were reviewed for cancer relevance; factors were either retained, dropped, or added to the adapted model on the basis of consensus in the study team and data availability for counties nationwide. County-level health factors were included that reflect intervenable targets for change in three categories: health infrastructure, physical environment, and social and economic factors (Table 1). Priority was placed on factors that reflect community conditions affected by policies and laws including health care delivery decisions rather than behaviors of individual community members. Weights were based on expert opinion and first assigned to the three categories using a similar top-down approach as for the PHI Model of Health.³ Health factors were generally perceived as having relatively modest (1%–2%), medium (4%–5%), or higher (8%) contributions to cancer risk within the health infrastructure (weighted 40%), physical environment (weighted 15%), and

TABLE 1. Cancer-Related Health Factors Included in the 2025 University of Wisconsin Population Health Institute Model of Health Adapted for Cancer

Health Factors	Weight, %	Data Source
Health infrastructure	40	
Access to exercise opportunities	4	ArcGIS Business Analyst and ArcGIS Online; YMCA; and US Census TIGER/Line Shapefiles from CHR&R ⁶
Food Environment Index	4	USDA Food Environment Atlas and Map the Meal Gap from Feeding America from CHR&R ⁶
Adult smoking	8	BRFSS from CHR&R ⁶
Adult obesity	4	BRFSS from CHR&R ⁶
Excessive drinking	2	BRFSS from CHR&R ⁶
Uninsured	4	Small area health insurance estimates from CHR&R ⁶
Primary care physicians (adjusted for county population)	4	Area Health Resource File/American Medical Association from CHR&R ⁶
Oncologists (adjusted for county population)	5	Centers for Medicare & Medicaid Services ⁷
Mammography screening	1	Mapping Medicare Disparities Tool from CHR&R ⁶
Colonoscopy screening	1	CDC State Cancer Profiles ⁵
Lung cancer screening	1	Mapping Medicare Disparities Tool ⁸
Prostate cancer screening	1	CDC State Cancer Profiles ⁵
Cervical cancer screening	1	CDC State Cancer Profiles ⁵
Physical environment	15	
Air pollution (PM2.5)	4	Environmental Public Health Tracking Network from CHR&R ⁶
Broadband Internet access	4	American Community Survey, 5-year estimates from CHR&R ⁶
Drinking water violations (no, yes)	1	Safe Drinking Water Information System data from CHR&R ⁶
Radon (zone 1-4)	1	EPA ⁹
Severe housing problems	4	CHAS data from CHR&R ⁶
Lead (DPI)	1	US Department of Housing and Urban Development ¹⁰
Social and economic factors	45	
High-school completion	8	American Community Survey, 5-year estimates from CHR&R ⁶
Some college	8	American Community Survey, 5-year estimates from CHR&R ⁶
Unemployment	8	Bureau of Labor Statistics from CHR&R ⁶
Children in poverty	8	Small Area Income and Poverty Estimates; and American Community Survey, 5-year estimates from CHR&R ⁶
Income inequality	8	American Community Survey, 5-year estimates from CHR&R ⁶
Social associations (per 100,000)	5	County Business Patterns from CHR&R ⁶

Abbreviations: BRFSS, Behavioral Risk Factor Surveillance System; CHAS, Comprehensive Housing Affordability Strategy; CHR&R, County Health Rankings & Roadmaps; DPI, deteriorated paint index; EPA, Environmental Protection Agency; GIS, geographic information system; PM, particulate matter; TIGER, Topological Integrated Geographic Encoding and Referencing Line; YMCA, Young Men's Christian Association.

social and economic factor categories (weighted 45%). Weights and selected measures were similar but not exactly the same as those in the PHI Model of Health as a result of adapting the model for cancer as an outcome.^{11,12}

Health infrastructure factors included measures related to health promotion and harm reductions (access to exercise opportunities, the Food Environment Index, smoking, obesity, and excessive drinking) and clinical care (percent uninsured under age 65 years, number of primary care physicians and oncologists adjusted for population size, and screening rates for breast, colorectal, lung, prostate, and cervical cancers). Physical environment factors included measures related to air, water, and land (particulate matter [PM] 2.5, radon levels, drinking water violations, and broadband Internet

access) and housing (severe housing problems and lead exposure). Finally, social and economic factors included measures for education (high-school graduation and completion of some college), income, employment, and wealth (unemployment, children in poverty, and income inequality) and social support (social associations).

Screening prevalence for colorectal, prostate, and cervical cancers were obtained from the CDC State Cancer Profiles website.⁵ Screening for colorectal cancer was defined as having completed a colonoscopy in the past 10 years for adults age 50–75 years. Lead exposure was determined via the deteriorated paint index, which reflects the number of pre-1980 households with large areas of deteriorated paint (a predictor of exposure to lead dust), obtained from the US

Department of Housing and Urban Development website.¹⁰ Radon exposure zones were obtained from the Environmental Protection Agency website.⁹ The number of oncologists for each county was obtained from the Centers for Medicare & Medicaid Services National Downloadable File.⁷ All other health factors and their data sources were obtained from the PHI website.⁶

Statistical Analysis

County-level z-scores were created for the health factors.¹³ For beneficial exposures (eg, cancer screening), values were inverted so that higher values were expected to increase cancer risk. Standardized values that were <−3 or >3 were truncated to −3 or 3, respectively.⁴ After standardization, weights were applied to the factors as listed in Table 1. The weighted values for the factors were summed to create a composite z-score for each county, with higher values indicating counties with less favorable health conditions.

K-means clustering of the composite z-scores was performed to group counties with similar health factor values, as described by Pollock et al.¹⁴ Clusters were identified numerically with higher clusters indicating less favorable county-level health factors. The total number of clusters was limited to 10 for ease of communication. As previously described, clustering counties with similar health conditions helps to mitigate error when ranking counties on the basis of composite z-scores that have unknown and varying levels of uncertainty.¹⁴

Associations between the composite z-score clusters (1–10) and cancer mortality rates at the county level were estimated using negative binomial regression models, with adjustment for county median age and including county population size

as an offset. Associations were reported as mortality rate ratios (RRs) with 95% CIs with counties in cluster 1 (most favorable health conditions) serving as the reference category. Furthermore, the age-standardized mortality rates per 100,000 population were calculated for each cluster.

The percentage of missing values for county-level health factors ranged from 0% to 5.0%. Missing values were imputed using the state-specific median value before data standardization and weighting. All analyses were completed using R version 4.3.1, including the “MASS” package for running negative binomial models and the “Ckmeans.1d.dp” package for K-means clustering.^{15–17}

To examine how health conditions varied across health factor clusters, four counties were randomly selected across regions in the United States in clusters 1, 4, 7, and 10 using a random number generator. For ease of demonstration, only health factors measured on the same scale (percentages) were plotted.

RESULTS

Across the 3,079 US counties with data, age-adjusted all-cancer mortality rates ranged from 48.9 (95% CI, 35.3 to 66.6; Mono County, CA) to 387.8 (95% CI, 348.9 to 430.1; Union County, FL) per 100,000 for 2018–2022. All-cancer age-adjusted mortality rates ranged from 126.2 per 100,000 for health factor cluster 1 (z-scores −1.53 to −0.72) to 165.5 per 100,000 for cluster 10 (RR, 1.49 [95% CI, 1.39 to 1.60]; z-scores 1.28 to 1.88; Table 2; Appendix Table A1).

Rate ratios for all-cancer mortality generally increased across health factor clusters (Fig 1, Table 2, Appendix Table A2). Comparing cluster 10 to 1, the rate ratio for lung cancer

TABLE 2. Cancer Mortality Rate Ratios According to Counties Clustered by Health Factors Included in the Adapted 2025 University of Wisconsin Population Health Institute Model of Health, United States (N = 3,079 Counties)

Health Factor Cluster	Z-Score Range	No. of Counties ^a	Mortality Rate Ratios (95% CI)		
			All Cancer Sites	Lung Cancer	Colorectal Cancer
1 (most healthy)	−1.53 to −0.72	194	1 (ref)	1 (ref)	1 (ref)
2	−0.72 to −0.46	442	1.16 (1.13 to 1.19)	1.26 (1.20 to 1.32)	1.13 (1.09 to 1.18)
3	−0.46 to −0.24	448	1.23 (1.20 to 1.26)	1.42 (1.35 to 1.49)	1.22 (1.18 to 1.27)
4	−0.24 to −0.03	540	1.31 (1.27 to 1.34)	1.60 (1.53 to 1.67)	1.33 (1.28 to 1.38)
5	−0.03 to 0.18	419	1.37 (1.33 to 1.40)	1.71 (1.63 to 1.79)	1.43 (1.37 to 1.49)
6	0.18 to 0.40	350	1.44 (1.40 to 1.48)	1.90 (1.80 to 2.00)	1.53 (1.46 to 1.60)
7	0.40 to 0.64	318	1.47 (1.42 to 1.51)	1.96 (1.86 to 2.07)	1.67 (1.58 to 1.76)
8	0.64 to 0.91	209	1.51 (1.46 to 1.56)	1.99 (1.88 to 2.12)	1.75 (1.63 to 1.87)
9	0.91 to 1.25	116	1.56 (1.49 to 1.62)	2.10 (1.94 to 2.27)	1.69 (1.54 to 1.86)
10 (least healthy)	1.28 to 1.88	43	1.49 (1.39 to 1.60)	1.75 (1.51 to 2.02)	2.10 (1.72 to 2.53)

NOTE. All cancer mortality rates are for the years 2018–2022.

Abbreviation: ref, reference category.

^aThe total number of counties in each cluster for all cancer sites combined. The number of counties for each outcome is all cancer—3,079; lung cancer—2,745; and colorectal cancer—2,180.

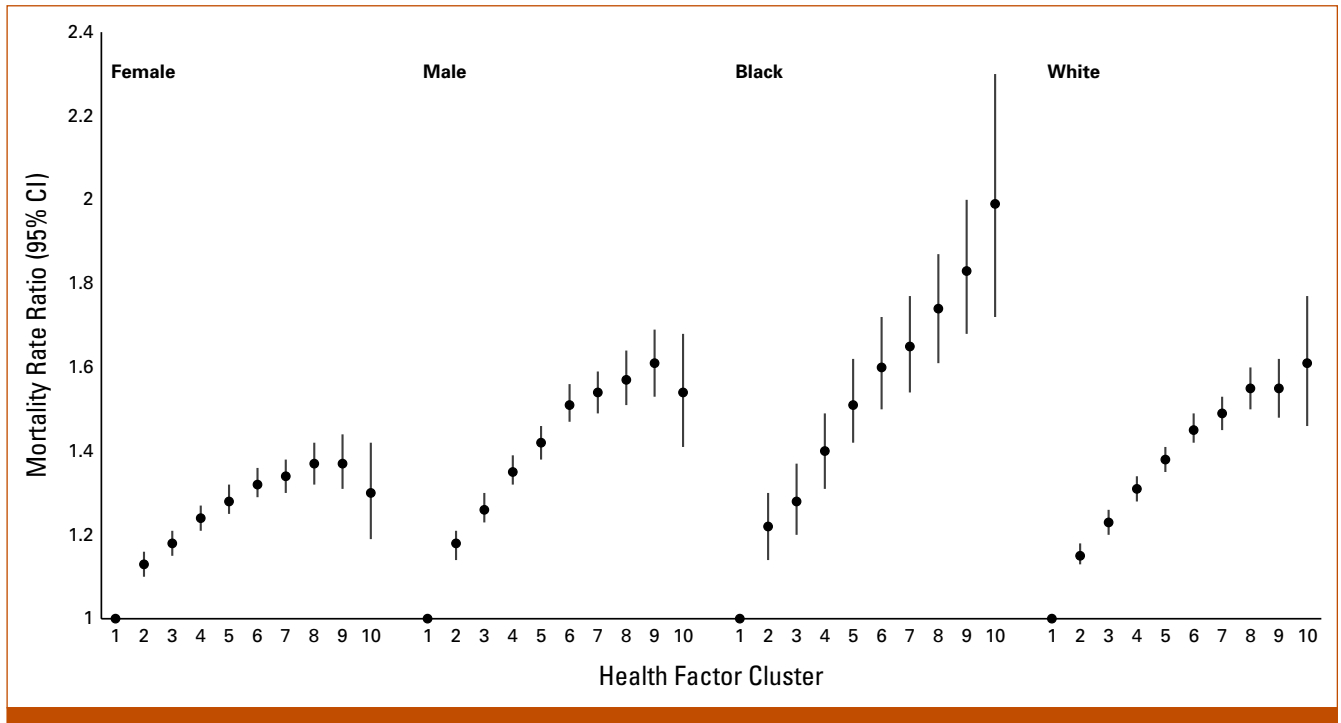


FIG 1. Age-adjusted rate ratios (and 95% CI) for all-cancer mortality according to health factor county cluster for female, male, Black, and White populations. Clusters range from 1 (most healthy community conditions, reference category) to 10 (least healthy).

(1.75 [95% CI, 1.51 to 2.02]) and colorectal cancer (2.10 [95% CI, 1.72 to 2.53]) was larger than for breast cancer (1.22 [95% CI, 0.90 to 1.60] among females) and prostate cancer (1.53 [95% CI, 1.06 to 2.13] among males).

Rate ratio estimates for all cancers, lung cancer, and colorectal cancer were higher for males than for females (Fig 1, Appendix Table A2). For example, comparing cluster 10 to 1, the rate ratio for lung cancer was 1.63 (95% CI, 1.26 to 2.08) for females and 1.97 (95% CI, 1.60 to 2.41) for males.

CIs were wider and rate ratio estimates were generally greater for the Black than for the White population (Fig 1, Appendix Table A3). For example, comparing cluster 10 to 1, the rate ratio for colorectal cancer was 2.72 (95% CI, 1.54 to 4.42) for Black persons and 2.16 (95% CI, 1.44 to 3.09) for White persons.

To demonstrate how health conditions varied across health factor clusters, four counties were randomly selected across regions in the United States in clusters 1, 4, 7, and 10. As could be expected, colon cancer screening rates decreased and the percentage of children in poverty increased from cluster 1 to 10 (Fig 2). Other health factors did not increase or decrease in an orderly manner across clusters. For example, cervical cancer screening was lower in a cluster 1 county (66%) than for counties in clusters 4 and 7 (76% and 81%, respectively), and attending some college was lower in a cluster 7 county (36%) than for a cluster 10 county (51%).

In a sensitivity analysis to examine the impact of choices of factors and weights assigned to the various health factors,

different weights were assigned to slightly different health factors arranged in four categories according to the 2014 County Health Rankings Model of Health, including health behaviors (30%), clinical care (20%), social and economic factors (40%), and physical environmental factors (10%; Appendix Table A4). Clustering resulted in fewer counties in clusters 1, 2, and 4 (Appendix Table A5). Comparing cluster 10 to 1, rate ratios for all cancers (1.60 [95% CI, 1.52 to 1.69]), lung cancer (2.06 [95% CI, 1.86 to 2.28]), and colorectal cancer (1.79 [95% CI, 1.58 to 2.01]) were similar to estimates in the primary analysis (1.49, 1.75, and 2.10, respectively, Table 2).

DISCUSSION

By applying the PHI Model of Health to cancer mortality, we showed that the model distinguishes between groups of counties on the basis of community conditions that are conducive to larger or smaller cancer mortality burdens. Mortality rates across clusters of counties were as much as two-fold different, demonstrating that the adapted PHI Model of Health was flexible to characterize variation in cancer mortality rates across counties for several different demographic groups and cancer types. The purpose of the model was not to control for confounding factors or model all possible causal factors related to cancer mortality; instead, health factors in the model were restricted to intervenable targets for change. A strength of this approach is its acknowledgment that the factors supportive of health are inextricably interrelated, consistent with a systems science framework²; the PHI Model of Health offers an

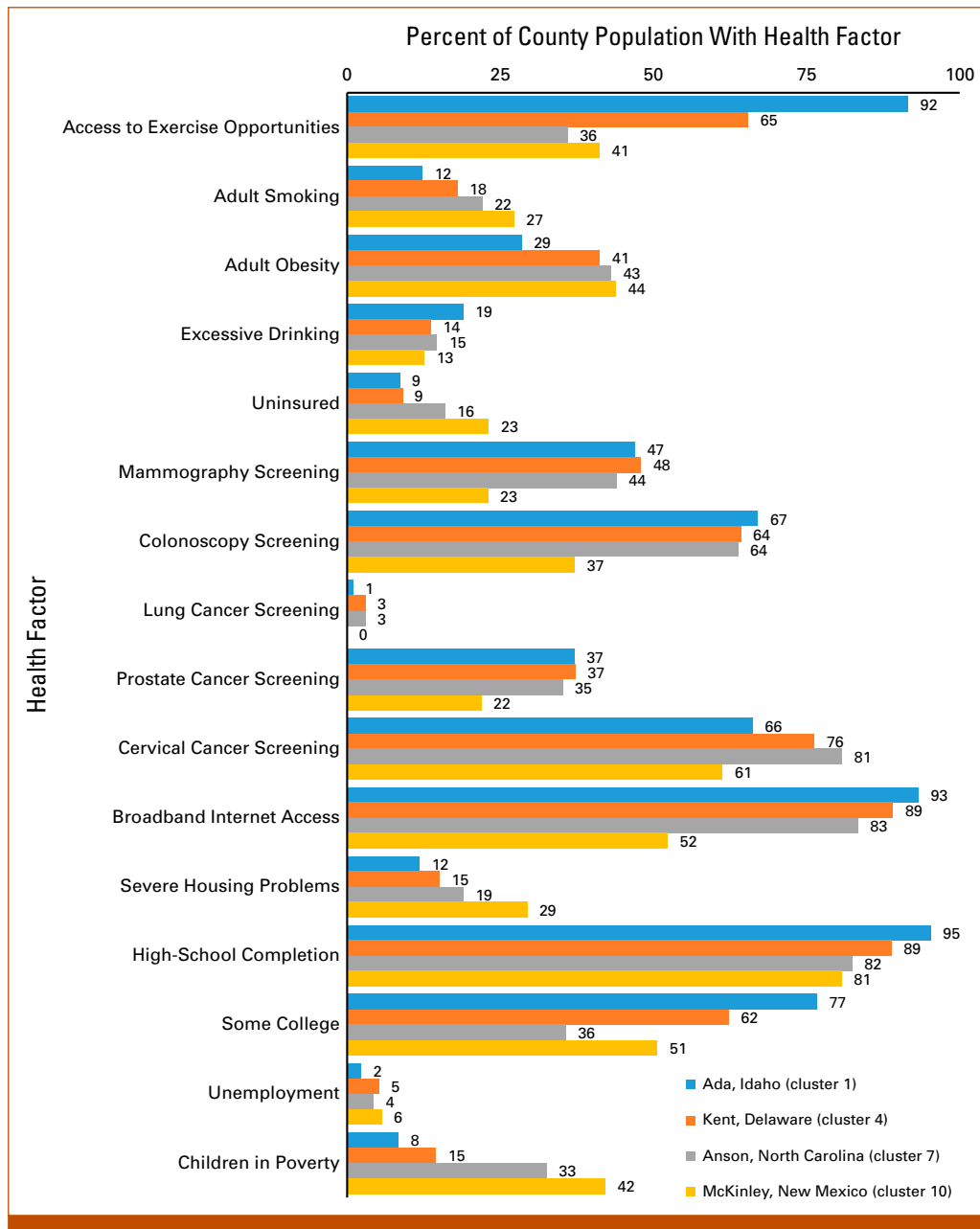


FIG 2. Histogram of selected health factors for four US counties. Counties randomly selected within clusters 1 (Ada County, Idaho), 4 (Kent County, Delaware), 7 (Anson County North Carolina), and 10 (McKinley County, New Mexico). Clusters represent community conditions that are most (cluster 1) to least (cluster 10) supportive of health. Counties have a range of population percent in rural areas⁶ (Ada 5.6%, Anson 77.8%, Kent 26.5%, and McKinley 66.5%) and age-adjusted all-cancer mortality rates per 100,000⁵ (Ada 131.8 [95% CI, 127.5 to 136.2]; Anson 191.2 [95% CI, 169.4 to 215.3]; Kent 168.1 [95% CI, 160.6 to 175.9]; and McKinley 143.0 [95% CI, 130.4 to 156.5]).

opportunity to identify intervenable priorities and provides an entry for the improvement of health factors within the local health infrastructure, physical environment, and social and economic ecosystems. Model results are intended to inform conversations and action about priorities for cancer prevention and control efforts, and should be complemented with knowledge of other important community conditions, such as rurality, and input of local residents.

This approach complements other techniques used by NCI-designated cancer centers to identify priorities for innovations in research, clinical care, outreach, and education. For many years, institutions have followed the example set by the American Cancer Society and generated lay-friendly reports that resemble Cancer Facts & Figures.¹⁸ More recently, dozens of centers are using Cancer InFocus, developed at the University of Kentucky Markey Cancer Center,¹⁹ which facilitates exploration of variations in cancer incidence and

mortality rates, demographic factors, and several other factors through maps. The Cancer InFocus website provides an interactive platform that complements static reports. Our approach uses much of the same data available on the Cancer InFocus website, with our methods also providing a way to integrate the multiple types of health factors into a single z-score. Numerous indices of health have been developed for similar purposes, including the Area Deprivation Index²⁰ and the Yost Index.²¹ However, these indices can include factors, such as home values, that lack evidence-based interventions for reducing the burden of diseases such as cancer.

The PHI Model of Health is proposed as a framework for identifying cancer priorities because, as mentioned immediately above, it is composed of intervenable targets for health improvement. Several resources are available to identify evidence-based interventions related to the factors in the model.²²⁻²⁴ For example, shared decision making between health care providers and patients for cancer screening and the provision of patient navigators when patients with cancer are first diagnosed are scientifically supported interventions.²⁴ Health factors in the PHI model are amenable to intervention at both the individual and population levels but call attention to the community conditions (social determinants of health) that require structural change at policy and systems levels, for example, tobacco taxes and expanded access to affordable health insurance. The PHI model is also flexible to identify geographic areas with the highest burden of all cancers combined or for specific cancer sites, and for specific demographic groups. The model can be applied to geographic units other than a county for which health factor and outcome data are available.

Several limitations of our approach indicate opportunities for improvement. Health factors weights are nominal and

not intended to directly reflect population attributable fractions. Our sensitivity analysis showed that cancer mortality rate ratios were elevated across the health factor clusters under a different set of weights. Using two sets of weights corresponding to the 2014 County Health Rankings and 2025 PHI Models of Health did not result in substantial differences in risk estimates or cluster definitions. However, national data were not available for some key cancer risk factors that ideally could be included such as human papillomavirus vaccination. Not all counties had sufficient data, and cancer mortality rates were suppressed for less populous counties and less common cancer types. County was used in this analysis and may represent the appropriate unit for intervention, for example, in a partnership with a county health department or other organizations that are based at the county level. County may not be the ideal geographic area for other partnerships or interventions and may also disguise heterogeneity of cancer risks within its borders. However, this approach provides a framework for straightforward application to geographic units other than county, for example, census tract, health service area, or province, for which relevant data are available.

The PHI has accumulated a track record of scholarship and coaching communities to implement positive changes for health and health equity since 2010. The framework described here provides an approach for identifying and linking community-specific conditions to the evidence-based interventions that hold promise to directly address drivers of the local cancer burden. This framework, together with input from community members and the scientific and clinical strengths of a cancer center, provides a data-driven model to prioritize efforts for reducing the cancer burden within cancer center catchment areas.

AFFILIATIONS

¹Carbone Cancer Center, School of Medicine and Public Health, University of Wisconsin-Madison, Madison, WI

²Department of Population Health Sciences, School of Medicine and Public Health, University of Wisconsin-Madison, Madison, WI

³Department of Biostatistics and Medical Informatics, School of Medicine and Public Health, University of Wisconsin-Madison, Madison, WI

⁴Department of Medicine, School of Medicine and Public Health, University of Wisconsin-Madison, Madison, WI

⁵Department of Family Medicine and Community Health, School of Medicine and Public Health, University of Wisconsin-Madison, Madison, WI

⁶Population Health Institute, School of Medicine and Public Health, University of Wisconsin-Madison, Madison, WI

CORRESPONDING AUTHOR

Amy Trentham-Dietz, PhD; e-mail: trentham@wisc.edu.

PRIOR PRESENTATION

Presented in part at the Catchment Area Data Excellence (CADEx) Conference, Coronado, CA, January 29-31, 2025; and The Annual

Meeting of the American Society of Preventive Oncology (ASPO), Philadelphia, PA, April 6-8, 2025.

SUPPORT

Supported by grant P30CA014520 by the National Institutes of Health/National Cancer Institute.

AUTHOR CONTRIBUTIONS

Conception and design: Amy Trentham-Dietz, Ronald E. Gangnon, Earlise C. Ward, Shaneda Warren Andersen, Marjory L. Givens

Financial support: Amy Trentham-Dietz

Administrative support: Allison R. Dahlke, Noelle K. LoConte

Collection and assembly of data: Amy Trentham-Dietz, Thomas P. Lawler, Earlise C. Ward, Christine P. Muganda

Data analysis and interpretation: Amy Trentham-Dietz, Thomas P. Lawler, Ronald E. Gangnon, Allison R. Dahlke, Noelle K. LoConte, Earlise C. Ward

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information

about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/cci/author-center.

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](http://OpenPayments.gov)).

Noelle K. LoConte

Consulting or Advisory Role: AbbVie, Personal Genome Diagnostics
Research Funding: Exact Sciences

No other potential conflicts of interest were reported.

REFERENCES

1. National Cancer Institute: National Institutes of Health: PAR-20-043: Cancer Center Support Grants (CCSGs) for NCI-designated cancer centers (P30 clinical trial optional). 2025. <https://grants.nih.gov/grants/guide/pa-files/PAR-21-321.html>
2. Luke DA, Powell BJ, Paniagua-Avila A: Bridges and mechanisms: Integrating systems science thinking into implementation research. *Annu Rev Public Health* 45:7-25, 2024
3. County Health Rankings & Roadmaps, University of Wisconsin Population Health Institute: County Health Rankings & Roadmaps Working Paper: Updates to methods and tools for practical application to improve health and equity. 2025. <https://www.countyhealthrankings.org/sites/default/files/media/document/Working%20Paper%20CHRR%202025.pdf>
4. Remington PL, Catlin BB, Gennuso KP: The County Health Rankings: Rationale and methods. *Popul Health Metr* 13:11, 2015
5. National Cancer Institute: National Institutes of Health, US Department of Health and Human Services: State Cancer Profiles. 2025. <https://statecancerprofiles.cancer.gov/>
6. County Health Rankings & Roadmaps: Health Data. 2025. <https://www.countyhealthrankings.org/health-data>
7. US Centers for Medicare & Medicaid Services: Doctors and Clinicians. 2025. <https://data.cms.gov/provider-data/topics/doctors-clinicians>
8. US Centers for Medicare & Medicaid Services: Mapping Medicare Disparities (MMD) Tool. 2024. <https://www.cms.gov/priorities/health-equity/minority-health/research-data/mapping-medicare-disparities-tool-mmd>
9. Environmental Protection Agency: The EPA Map of Radon Zones. 2024. <https://www.epa.gov/radon/epa-map-radon-zones>
10. Office of Policy Development and Research (PDMR): US Department of Housing and Urban Development: Deteriorated Paint Index by County. 2025. <https://hudgis-hud.opendata.arcgis.com/maps/HUD:deteriorated-paint-index-by-county>
11. Hood CM, Gennuso KP, Swain GR, et al: County health rankings: Relationships between determinant factors and health outcomes. *Am J Prev Med* 50:129-135, 2016
12. Park H, Roubal AM, Jovaag A, et al: Relative contributions of a set of health factors to selected health outcomes. *Am J Prev Med* 49:961-969, 2015
13. County Health Rankings & Roadmaps: Methods. 2025. <https://www.countyhealthrankings.org/health-data/methodology-and-sources/methods>
14. Pollock EA, Gangnon RE, Gennuso KP, et al: Cluster analysis methods to support population health improvement among US counties. *J Public Health Manag Pract* 30:E319-E328, 2024
15. R Core Team: R: A Language and Environment for Statistical Computing. Vienna, Austria, R Foundation for Statistical Computing, 2025. <https://www.R-project.org/>
16. Venables WN, Ripley BD: *Modern Applied Statistics With S* (ed 4). New York, NY, Springer, 2002. ISBN 0-387-95457-0
17. Wang H, Song M: Ckmeans.1d.dp: Optimal k-means clustering in one dimension by dynamic programming. *R J* 3:29-33, 2011
18. American Cancer Society: Cancer Facts & Figures 2025. Atlanta, GA, American Cancer Society, 2025. <https://www.cancer.org/research/cancer-facts-statistics.html>
19. Burus JT, Park L, McAfee CR, et al: Cancer InFocus: Tools for cancer Center catchment area geographic data collection and visualization. *Cancer Epidemiol Biomarkers Prev* 32:OF1-OF5, 2023
20. Kind AJH, Buckingham WR: Making neighborhood-disadvantage metrics accessible - The neighborhood Atlas. *N Engl J Med* 378:2456-2458, 2018
21. Yost K, Perkins C, Cohen R, et al: Socioeconomic status and breast cancer incidence in California for different race/ethnic groups. *Cancer Causes Control* 12:703-711, 2001
22. National Cancer Institute: National Institutes of Health: Evidence-Based Cancer Control Programs (EGCCP). 2024. <https://ebccp.cancercontrol.cancer.gov/index.do>
23. Office of Disease Prevention and Health Promotion, Office of the Assistant Secretary for Health, Office of the Secretary, et al: Healthy people 2030: Tools for action. 2025. <https://odphp.health.gov/healthypeople/tools-action>
24. County Health Rankings & Roadmaps: Strategies and solutions. 2025. <https://www.countyhealthrankings.org/strategies-and-solutions>

APPENDIX

TABLE A1. Cancer Mortality Rates According to Counties Clustered by Health Factors Included in the Adapted 2025 University of Wisconsin Population Health Institute Model of Health, United States (N = 3,079 counties)

Health Factor Cluster ^a	Z-Score Range	No. of Counties ^a	All Cancer	Lung	Colorectal	Breast (female)	Prostate (male)
1 (most healthy)	−1.53 to −0.72	194	126.2	23.7	10.5	17.2	17.3
2	−0.72 to −0.46	442	136.8	27.8	11.5	18.2	18.3
3	−0.46 to −0.24	448	143.1	30.4	12.1	18.9	18.2
4	−0.24 to −0.03	540	150.5	34.0	13.0	19.6	18.2
5	−0.03 to 0.18	419	148.7	29.9	13.7	20.0	19.6
6	0.18 to 0.40	350	154.0	33.2	14.3	20.8	19.4
7	0.40 to 0.64	318	165.3	36.8	15.6	21.2	20.5
8	0.64 to 0.91	209	170.1	37.0	16.4	22.4	22.0
9	0.91 to 1.25	116	144.4	26.5	13.6	19.4	18.5
10 (least healthy)	1.28 to 1.88	43	165.5	24.2	19.2	19.1	21.1

NOTE. Data presented as the mean of the county-level age-standardized mortality rates for the years 2018-2022 weighted by the inverse of the squared SE.

^aThe total number of counties in each cluster for all cancer sites combined. The number of counties for each outcome is all cancer—3,079; lung cancer—2,745; colorectal cancer—2,180; breast cancer—1,778; and prostate cancer—1,612.

TABLE A2. Cancer Mortality Rate Ratios According to Counties Clustered by Health Factors Included in the Adapted 2025 University of Wisconsin Population Health Institute Model of Health by Sex, United States

Health Factor Cluster			Mortality Rate Ratios (95% CI)			
Female Population	Z-Score Range	No. of Counties ^a	All Cancer Sites	Lung Cancer	Colorectal Cancer	Breast Cancer
1 (most healthy)	−1.53 to −0.72	190	1 (ref)	1 (ref)	1 (ref)	1 (ref)
2	−0.72 to −0.46	416	1.13 (1.10 to 1.16)	1.21 (1.14 to 1.27)	1.10 (1.05 to 1.16)	1.08 (1.04 to 1.12)
3	−0.46 to −0.24	428	1.18 (1.15 to 1.21)	1.33 (1.27 to 1.41)	1.17 (1.11 to 1.22)	1.13 (1.09 to 1.17)
4	−0.24 to −0.03	514	1.24 (1.21 to 1.27)	1.46 (1.38 to 1.53)	1.25 (1.19 to 1.32)	1.18 (1.13 to 1.22)
5	−0.03 to 0.18	408	1.28 (1.25 to 1.32)	1.51 (1.43 to 1.60)	1.31 (1.24 to 1.38)	1.20 (1.15 to 1.25)
6	0.18 to 0.40	345	1.32 (1.29 to 1.36)	1.63 (1.53 to 1.72)	1.34 (1.25 to 1.42)	1.23 (1.17 to 1.29)
7	0.40 to 0.64	307	1.34 (1.30 to 1.38)	1.64 (1.53 to 1.75)	1.49 (1.36 to 1.62)	1.31 (1.22 to 1.39)
8	0.64 to 0.91	203	1.37 (1.32 to 1.42)	1.64 (1.52 to 1.78)	1.60 (1.42 to 1.79)	1.31 (1.21 to 1.43)
9	0.91 to 1.25	112	1.37 (1.31 to 1.44)	1.66 (1.49 to 1.85)	1.31 (1.14 to 1.49)	1.22 (1.10 to 1.35)
10 (least healthy)	1.28 to 1.88	38	1.30 (1.19 to 1.42)	1.63 (1.26 to 2.08)	1.78 (1.23 to 2.48)	1.22 (0.90 to 1.60)

Male Population			All Cancer Sites	Lung Cancer	Colorectal Cancer	Prostate Cancer
1 (most healthy)	−1.53 to −0.72	191	1 (ref)	1 (ref)	1 (ref)	1 (ref)
2	−0.72 to −0.46	426	1.18 (1.14 to 1.21)	1.29 (1.22 to 1.36)	1.15 (1.09 to 1.21)	1.13 (1.07 to 1.18)
3	−0.46 to −0.24	439	1.26 (1.23 to 1.30)	1.48 (1.40 to 1.57)	1.27 (1.21 to 1.33)	1.18 (1.13 to 1.24)
4	−0.24 to −0.03	525	1.35 (1.32 to 1.39)	1.71 (1.63 to 1.81)	1.39 (1.32 to 1.46)	1.19 (1.13 to 1.25)
5	−0.03 to 0.18	411	1.42 (1.38 to 1.46)	1.85 (1.75 to 1.95)	1.49 (1.42 to 1.57)	1.26 (1.20 to 1.33)
6	0.18 to 0.40	348	1.51 (1.47 to 1.56)	2.11 (1.98 to 2.23)	1.60 (1.50 to 1.70)	1.25 (1.17 to 1.34)
7	0.40 to 0.64	310	1.54 (1.49 to 1.59)	2.21 (2.07 to 2.35)	1.83 (1.70 to 1.97)	1.36 (1.26 to 1.47)
8	0.64 to 0.91	203	1.57 (1.51 to 1.64)	2.26 (2.10 to 2.43)	1.89 (1.71 to 2.09)	1.45 (1.30 to 1.61)
9	0.91 to 1.25	113	1.61 (1.53 to 1.69)	2.34 (2.13 to 2.58)	1.72 (1.51 to 1.95)	1.39 (1.20 to 1.59)
10 (least healthy)	1.28 to 1.88	41	1.54 (1.41 to 1.68)	1.97 (1.60 to 2.41)	2.23 (1.62 to 2.97)	1.53 (1.06 to 2.13)

NOTE. All cancer mortality rates are for the years 2018-2022.

^aThe total number of counties in each cluster. The number of counties for each outcome is all cancer—3,007 (males), 2,961 (females); lung cancer—2,466 (males), 2,269 (females); colorectal cancer—1,585 (males), 1,368 (females); breast cancer—1,778 (females); and prostate cancer—1,612 (males).

TABLE A3. Cancer Mortality Rate Ratios According to Counties Clustered by Health Factors Included in the Adapted 2025 University of Wisconsin Population Health Institute Model of Health, Black and White Population, United States

Cluster	Z-Score Range	No. of Counties ^a	All Cancer (95% CI)	Lung (95% CI)	Colorectal (95% CI)	Breast (female; 95% CI)	Prostate (male; 95% CI)
Black population							
1 (most healthy)	−1.53 to −0.72	88	1 (ref)	1 (ref)	1 (ref)	1 (ref)	1 (ref)
2	−0.72 to −0.46	127	1.22 (1.14 to 1.30)	1.36 (1.19 to 1.55)	1.19 (1.05 to 1.36)	1.15 (1.01 to 1.30)	1.28 (1.09 to 1.50)
3	−0.46 to −0.24	148	1.28 (1.20 to 1.37)	1.45 (1.28 to 1.65)	1.18 (1.04 to 1.33)	1.12 (1.00 to 1.26)	1.30 (1.11 to 1.51)
4	−0.24 to −0.03	192	1.40 (1.31 to 1.49)	1.62 (1.43 to 1.83)	1.36 (1.20 to 1.53)	1.20 (1.07 to 1.35)	1.42 (1.22 to 1.66)
5	−0.03 to 0.18	159	1.51 (1.42 to 1.62)	1.75 (1.54 to 1.99)	1.47 (1.30 to 1.66)	1.29 (1.15 to 1.45)	1.53 (1.31 to 1.79)
6	0.18 to 0.40	143	1.60 (1.50 to 1.72)	1.99 (1.73 to 2.29)	1.46 (1.28 to 1.67)	1.25 (1.10 to 1.42)	1.45 (1.21 to 1.73)
7	0.40 to 0.64	159	1.65 (1.54 to 1.77)	2.14 (1.86 to 2.46)	1.76 (1.48 to 2.08)	1.32 (1.09 to 1.59)	1.63 (1.27 to 2.06)
8	0.64 to 0.91	113	1.74 (1.61 to 1.87)	2.32 (1.98 to 2.71)	2.03 (1.61 to 2.52)	1.33 (0.97 to 1.79)	1.89 (1.38 to 2.55)
9	0.91 to 1.25	68	1.83 (1.68 to 2.00)	2.47 (2.05 to 2.97)	1.44 (1.14 to 1.80)	1.29 (1.04 to 1.60)	1.50 (1.08 to 2.06)
10 (least healthy)	1.28 to 1.88	19	1.99 (1.72 to 2.30)	2.49 (1.76 to 3.46)	2.72 (1.54 to 4.42)	2.01 (1.07 to 3.42)	2.56 (1.14 to 4.98)
White population							
1 (most healthy)	−1.53 to −0.72	193	1 (ref)	1 (ref)	1 (ref)	1 (ref)	1 (ref)
2	−0.72 to −0.46	439	1.15 (1.13 to 1.18)	1.25 (1.20 to 1.30)	1.12 (1.08 to 1.17)	1.07 (1.02 to 1.11)	1.08 (1.03 to 1.14)
3	−0.46 to −0.24	444	1.23 (1.20 to 1.26)	1.44 (1.38 to 1.50)	1.22 (1.17 to 1.27)	1.09 (1.05 to 1.13)	1.11 (1.06 to 1.17)
4	−0.24 to −0.03	538	1.31 (1.28 to 1.34)	1.64 (1.57 to 1.70)	1.32 (1.27 to 1.37)	1.14 (1.10 to 1.19)	1.10 (1.04 to 1.16)
5	−0.03 to 0.18	418	1.38 (1.35 to 1.41)	1.76 (1.69 to 1.84)	1.43 (1.37 to 1.49)	1.20 (1.15 to 1.26)	1.20 (1.13 to 1.26)
6	0.18 to 0.40	349	1.45 (1.42 to 1.49)	1.97 (1.88 to 2.06)	1.50 (1.42 to 1.57)	1.22 (1.15 to 1.28)	1.13 (1.05 to 1.22)
7	0.40 to 0.64	309	1.49 (1.45 to 1.53)	2.07 (1.97 to 2.17)	1.65 (1.55 to 1.76)	1.31 (1.21 to 1.42)	1.32 (1.19 to 1.45)
8	0.64 to 0.91	204	1.55 (1.50 to 1.60)	2.20 (2.08 to 2.33)	1.79 (1.65 to 1.95)	1.34 (1.20 to 1.50)	1.45 (1.24 to 1.69)
9	0.91 to 1.25	107	1.55 (1.48 to 1.62)	2.24 (2.07 to 2.43)	1.84 (1.61 to 2.09)	1.32 (1.09 to 1.59)	1.33 (1.02 to 1.69)
10 (least healthy)	1.28 to 1.88	31	1.61 (1.46 to 1.77)	2.51 (2.07 to 3.01)	2.16 (1.44 to 3.09)	1.03 (0.26 to 2.68) ^b	^c

NOTE. All cancer mortality rates are for the years 2018-2022.

^aThe total number of counties in each cluster for all cancer sites combined. For the White population, the number of counties for each outcome is all cancer—1,216; lung cancer—665; colorectal cancer—386; breast cancer—341; and prostate cancer—300. For the Black population, the number of counties for each outcome is all cancer—3,032; lung cancer—2,654; colorectal cancer—1,970; breast cancer—1,599; and prostate cancer—1,425.

^bOnly one county had at least three breast cancer deaths among White women in cluster 10.

^cZero counties in cluster 10 have at least three prostate cancer deaths per year on average among White men.

TABLE A4. Cancer-Related Health Factors Included in the 2014 County Health Rankings Model of Health Adapted for Cancer

Health Factors	Weight, %	Data Source
Health behaviors	30	
Adult smoking	10	BRFSS from CHR&R ⁶
Adult obesity	10	BRFSS from CHR&R ⁶
Physical inactivity	3	BRFSS from CHR&R ⁶
Sunburn risk (UV-B, Wh/m ²)	2	National Cancer Institute ^a
Excessive drinking	3	BRFSS from CHR&R ⁶
Hepatitis C mortality (per 100,000)	1	HepVu ^b
HIV prevalence (per 100,000)	1	From CHR&R ⁶
Clinical care	20	
Uninsured	5	Small area health insurance estimates from CHR&R ⁶
Primary care physicians (ratio)	2	Area Health Resource File/American Medical Association from CHR&R ⁶
Dentists (ratio)	1	Area Health Resource File/National Provider Identifier Downloadable File from CHR&R ⁶
Oncologists (per county)	1	Centers for Medicare & Medicaid Services ⁷
Broadband Internet access	1	American Community Survey, 5-year estimates from CHR&R website ⁶
Mammography screening	2	Mapping Medicare Disparities Tool from CHR&R ⁶
Colonoscopy screening	2	CDC State Cancer Profiles ⁵
Lung cancer screening	2	Mapping Medicare Disparities Tool ⁸
Prostate cancer screening	2	CDC State Cancer Profiles ⁵
Cervical cancer screening	2	CDC State Cancer Profiles ⁵
Physical environment	10	
Air pollution (PM2.5)	2	Environmental Public Health Tracking Network from CHR&R ⁶
Radon (zone 1-4)	2	EPA ⁹
Lead (DPI)	1	US Department of Housing and Urban Development ¹⁰
Access to exercise opportunities	2.5	ArcGIS Business Analyst and ArcGIS Online; YMCA; and US Census TIGER/Line Shapefiles from CHR&R ⁶
Food Environment Index	2.5	USDA Food Environment Atlas and Map the Meal Gap from Feeding America from CHR&R ⁶
Social and economic factors	40	
High-school completion	5	American Community Survey, 5-year estimates from CHR&R ⁶
Some college	5	American Community Survey, 5-year estimates from CHR&R ⁶
Unemployment	10	Bureau of Labor Statistics from CHR&R ⁶
Children in poverty	7.5	Small Area Income and Poverty Estimates; and American Community Survey, 5-year estimates from CHR&R ⁶
Income inequality	2.5	American Community Survey, 5-year estimates from CHR&R ⁶
Social associations (per 100,000)	5	County Business Patterns from CHR&R ⁶
Black/White residential segregation (Index of Dissimilarity)	5	From CHR&R ⁶

Abbreviations: CDC, Centers for Disease Control; CHR, County Health Rankings; CMS, Centers for Medicare & Medicaid Services; DPI, deteriorated paint index; EPA, Environmental Protection Agency; PM, particulate matter.

^aGIS Portal for Cancer Research, National Cancer Institute, National Institutes of Health: County Level UV Exposure Data for the Continental United States. 2025. <https://gis.cancer.gov/tools/uv-exposure/>.

^bHepVu: HepVu-Mapping the Hepatitis Epidemic. <https://hepvu.org/>.

TABLE A5. Cancer Mortality Rate Ratios According to Counties Clustered by Health Factors Included in the Adapted 2014 County Health Rankings Model of Health, United States (N = 3,079 counties)

Health Factor Cluster ^a	Z-Score Range	No. of Counties ^b	All Cancer (95% CI)	Lung (95% CI)	Colorectal (95% CI)	Breast (female; 95% CI)	Prostate (male; 95% CI)
1 (most healthy)	−1.40 to −0.82	109	1 (ref)	1 (ref)	1 (ref)	1 (ref)	1 (ref)
2	−0.81 to −0.54	278	1.14 (1.10 to 1.17)	1.22 (1.16 to 1.29)	1.10 (1.05 to 1.15)	1.08 (1.03 to 1.12)	1.10 (1.04 to 1.16)
3	−0.54 to −0.32	464	1.26 (1.22 to 1.29)	1.46 (1.38 to 1.54)	1.22 (1.17 to 1.27)	1.15 (1.10 to 1.20)	1.19 (1.13 to 1.26)
4	−0.32 to −0.12	460	1.33 (1.29 to 1.37)	1.64 (1.56 to 1.73)	1.31 (1.25 to 1.37)	1.15 (1.11 to 1.20)	1.20 (1.14 to 1.27)
5	−0.12 to 0.07	493	1.39 (1.35 to 1.43)	1.76 (1.67 to 1.85)	1.40 (1.35 to 1.47)	1.22 (1.17 to 1.27)	1.25 (1.18 to 1.32)
6	0.08 to 0.27	419	1.48 (1.44 to 1.52)	1.96 (1.85 to 2.06)	1.54 (1.47 to 1.61)	1.29 (1.23 to 1.35)	1.31 (1.23 to 1.39)
7	0.27 to 0.48	351	1.52 (1.48 to 1.57)	2.09 (1.98 to 2.21)	1.61 (1.53 to 1.70)	1.26 (1.19 to 1.33)	1.25 (1.16 to 1.35)
8	0.48 to 0.72	266	1.53 (1.48 to 1.58)	2.09 (1.97 to 2.22)	1.65 (1.56 to 1.76)	1.27 (1.19 to 1.36)	1.39 (1.28 to 1.52)
9	0.72 to 1.03	159	1.63 (1.57 to 1.70)	2.28 (2.12 to 2.44)	1.93 (1.77 to 2.09)	1.42 (1.28 to 1.58)	1.55 (1.35 to 1.78)
10 (least healthy)	1.04 to 1.76	80	1.60 (1.52 to 1.69)	2.06 (1.86 to 2.28)	1.79 (1.58 to 2.01)	1.28 (1.11 to 1.46)	1.34 (1.11 to 1.61)

NOTE. All cancer mortality rates are for the years 2018-2022.

^aClusters are arranged in order from most to least healthy characteristics.

^bThe total number of counties in each cluster for all cancer sites combined. The number of counties for each outcome is: all cancer—3,079; lung cancer—2,745; colorectal cancer—2,180; breast cancer—1,778; and prostate cancer—1,612.